# Paul Leger, Ph. D. in Computer Science & Software Engineer

Software Engineer & Associate Professor
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## Personal

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| Name: | Paul Leger |
| Hometown: | Coquimbo, Chile |
| Nationality: | Chilean |
| **Summary:** | |
| Google Scholar: | CrqCdsgAAAAJ |
| ORCID: | 0000-0003-0969-5139 |
| SCOPUS: | 36087550600 |
| DBLP: | https://dblp.org/pid/78/7936.html |
| Directors/Headships: | 6 Positions |
| Indexed Journals: | 39 Publications (34 WoS-JCR, 2 SCOPUS, and 3 Latindex) |
| Indexed Conferences: | 19 Publications (SCOPUS) |
| Indexed Workshops: | 8 Publications (SCOPUS) |
| Journal Editions: | 6 Journals |
| First/Corresp Author: | 25 (represents 37%) |
| Courses: | 59 Courses (52 undergrad and 7 graduate) |
| Thesis Supervisions: | 29 Theses (26 undergraduate and 3 graduate) |
| Relevant Software: | 9 Developments (2 applications, 3 libraries, and 4 languages) |
| External Projects: | 3 with funding |



## Education

**Visiting Researcher**, universities across the world:
- Centre for Development of Advanced Computing (C-DAC), India, 2024 (3 weeks).
- University of The Andes, Colombia, 2021 (3 weeks).
- Tokyo Institute of Technology, Japan, 2019 (6 months).
- University of Cauca, Colombia, 2018 (2 months).
- Shibaura Institute of Technology ($x$ 2), Japan, 2013-2014 (4-2 months).
- École des mines de Nantes, France, 2010 (2 months).

**Certifications:** *a*) "Fundamentals of English Medium Instruction", University of Pennsylvania, 2023, *b*) "Teaching University", Highest level, Catholic University of the North, 2024.

**Ph. D. in Computer Science**, University of Chile, 2006 - 2012.
- Subject: Opening Up Trace-Based Mechanisms.
- Advisor: Éric Tanter (https://pleiad.cl/etanter).

**Engineering degree in Computer Science**, University of La Serena, 1999 - 2005 (July).

**B.Sc. in Computing Science**, University of La Serena, 1999 - 2005 (April).

## Management Services

**Director.** Engineering School (https://eic.ucn.cl) at Universidad Católica del Norte (Chile). Management of a budget for a four-floor engineering school with 1,000+ students, guiding professors, and a teaching support unit, 2024 - 2026.

**Director.** Chilean Computer Science Society (SCCC), 2020 - 2022.

**Deputy Director.** Engineering Systems Doctorate (https://dsi-en.ucn.cl) at Universidad Católica del Norte (Chile), 2023 - 2024.

**Head of Program Studies.** 1) Civil Engineering in Computer Science and 2) Information Technology Engineering at the Universidad Católica del Norte (Chile), over 400 students, 2021 - 2022.

**Pragmatics Lab.** Director of an international research laboratory (http://pragmaticslab.com), where professors and students from Japan, Colombia, and Chile collaborate, 2018 - present.

## Software Development Engineer

As I enjoy working on software engineering, I occasionally work as a freelancer doing software development and as a team manager in software projects:

**Macrosoftware.** Consulting software engineer, 2020 - present.

**G-Tech.** Project Manager, a startup mobile application, 2022.

**Autopista Central.** Consulting engineer, 2017.

**Regional Observatory (working market).** Software project manager, implementation of an information system that allows users to analyze working market in a region of Chile, 2016 - 2017.

**Autopista Central.** Consulting engineer, 2011.

## Software Development for Research

Research-driven software systems and developer tools, built with production-grade engineering practices where applicable (open-source repos, demos, and reusable libraries). Source: https://pleger.cl/software.

**PL/SE research with AI:** Implementation with many upgrades of my previous research in Programming Language and Software Engineering with Artificial Intelligence agents (Codex): AspectScript, Expressive Stateful Aspect (ESA-myPhD), DeloreanGo (adaptation from DeloreanJs), Sync/CC, and EMA-Kotlin (adaptation from EMA-js). https://pleger.github.io/PL-SE-research-with-AI, 2026. [**ARTIFICIAL INTELLIGENCE**].

**Fakenews-ABM:** ABM for the strategy evaluations for fake news dissemination on social network sites. https://github.com/pleger/FAKENEWS-ABM-News-Source-Strategies, 2026. [**AGENT-BASED MODELS**].

**SBABM:** Java software to simulate buyers in e-commerce like Amazon, AliExpress, and Wish. An agent-based model to simulate the e-commerce behavior between agents and markets. https://github.com/pragmaticslaboratory/SBABM, 2023. [**APPLICATION**].

**EMAjs:** JavaScript library that implements an Expressive and Modular Activation mechanism for context-oriented programming. https://github.com/pragmaticslaboratory/EMAjs, 2022. [**LANGUAGE**].

**DeloreanJS:** A back-in-time debugger for JavaScript. This software is strongly developed by Felipe Ruiz (felipe.ruiz@alumnos.ucn.cl) and Guillermo Victorero (guillermo.victorero@alumnos.ucn.cl). You can try it on https://pleger.cl/sites/deloreanjs. https://github.com/pragmaticslaboratory/deloreanjs, 2019. [**APPLICATION**].

**AspectScript:** An aspect-oriented extension of JavaScript for expressive AOP, which integrates several state-of-the-art features related to higher-order programming and expressive scoping of aspects. https://pleiad.cl/aspectscript, 2010. [**LANGUAGE**].

**FASOW:** FASOW provides an Open Implementation for Agent-Based Models applied to e-WOM Marketing Campaigns. This software is strongly developed by Diego Nuñez (diego.nunez.trabajo@gmail.com). https://github.com/pragmaticslaboratory/fasow, 2024. [**LIBRARY**].

**Matcher Cell:** A JavaScript library to flexibly match patterns. You can try it on https://pragmaticslaboratory.github.io/matcher-cells-study-cases. https://github.com/pragmaticslaboratory/match-cell-base, 2021. [**LANGUAGE**].

**Sync/CC (proof of concept):** A library to address callback issues on JavaScript. https://github.com/pragmaticslaboratory/synnccc, 2019. [**LIBRARY**].

## Academic Experience

### *Scientific Services*

**Funded Projects** (*x 4*). Principal Researcher (ANID-2023), Principal Investigator (Chile), and Co-Researcher (Japanese government).

**Organizer of International Events** (*x 4*). JCC'20 (Chile), COP'20 (Germany), and MPSE'14/15 (USA).

**Editor and Reviewer of JCR Journals** (*x 50*). Editor (*x 5*) and Reviewer (*x 45*).

**Chair of International Events** (*x 9*). ICSME'23 (Colombia), LMN'22 (Chile), JCC'20 (Chile), MPSE'14/15 (USA), SAC-SE'17 (Morocco), and BIONETICS'15 (USA).

**Program Committee Member** (*x 24*). VMIL (Singapore), ICSME (various countries), SAC-SE (various countries), SBLP (Brazil), COP (Germany), VISSOFT (Chile), INFONOR (Chile), etc.

**Evaluator** (*x 16*). Research Projects and (International) External Master Reviewers.

## References

**Hidehiko Masuhara**. Ph. D. in Computer Science. Full Professor & Dean. Department of Mathematical and Computing Science. Tokyo Institute of Technology, Japan (masuhara@acm.org). Webpage: https://prg.is.titech.ac.jp/people/masuhara.

**Éric Tanter**. Ph. D. in Computer Science. Full Professor. Computer Science Department (DCC), University of Chile, Chile (etanter@dcc.uchile.cl). Webpage: https://pleiad.cl/people/etanter.

**Alexandre Bergel**. Ph. D. in Computer Science. Computer Scientist. Rational AI (https://relational.ai), Switzerland (alexandre.bergel@me.com). Webpage: https://bergel.eu.